



VEHICLE PARKING

Efficient parking platform

A PROJECT REPORT



VEHICLE PARKING APPLICATION

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Name: SHIVAM KUMAR

Email: 23f1003137@ds.study.iitm.ac.in I'm currently pursuing the IITM BS program in Data Science and Programming. I'm super passionate about tech and love building things—whether it's web apps, data-driven projects, or just experimenting with new ideas. I'm always up for a challenge and enjoy learning something new along the way. Excited about connecting with like-minded people and exploring the endless possibilities in tech.x

Description:

This project is built as part of the Modern Application Development - I course from the IITM BS Degree Program.

It is a 4-wheeler vehicle parking platform designed for both administrators and end-users to streamline the parking process with digital precision and convenience.

- *Admins have complete control over the platform. They can create, edit, and manage parking lots, define parking spots, view user activity, and monitor parking Users (vehicle owners) can view available lots, reserve parking spots, and track parking history. Real-time timestamp tracking ensures transparency in spot usage and summaries.*
- *release, along with cost of 1 hours by default and after 1hours calculations based on the duration of usage.*

The platform is developed with scalability and modularity in mind, allowing future enhancements such as smart parking recommendations, vehicle recognition, and IoT-based real-time monitoring.

Technologies used

❖ Backend:

- Flask: Web application framework
- Flask-SQLAlchemy: ORM for database interactions
- Flask-Login: Session and user management
- SQLite: Lightweight relational database
- Python Dotenv: For environment variables

❖ Frontend:

- HTML/CSS: Core structure and styling
- Bootstrap 5: Responsive components and layout
- Jinja2: Template rendering

❖ Tools and IDE:

- VS Code, SQLite Editor

FEATURES

- Admin/User login and role-based access
- Create, update, delete parking lots and spots
- Reserve and release parking spots
- Track booking history
- Dashboard view for admins and users
- Spot availability status (Available/Occupied)
- Search functionality for users, lots, and spots
- Cost calculation based on parking duration

DATABASE DESIGN

- The application uses a normalized relational database with foreign key constraints. Key tables include:

1. Users - id (PK) , fullname , email , password , pincode, role (user/admin)
2. Parking Lots - id (PK) , Lot_name , address , pin_code , price , total_spots
3. Parking Spots - id (PK) , lot_id (FK) , status (A/O), apot_number
4. Booking - id (PK), spot_id (FK) , user_id (FK) , lot_id (FK) ,booking_time, release_time , status , vehicle_number

APPLICATION STRUCTURE

vehicle_parking_app/

app.py # Main application starter and All Flask routes and controllers

templates # HTML templates

models.py # SQLAlchemy models (assumed) .

.venv # Environment variables

requirements.txt # Dependency list

ReadMe.md # Application description

WORKFLOW

1. UserSignup/Login
2. Admin Dashboard - Add/Edit/Delete parking lots, View and manage users , View all booking
3. User Dashboard - View available lots , Book a spot , Release a spot , View history

DEPLOYMENT AND TESTING

-Local deployment on port 5000 using python app.py

CONCLUSION

TheVehicle Parking App demonstrates the use of Flask, database modeling, and user-session handling in building a real-world web application. It provides a modular, extendable solution for smart parking in urban environments.

video:- <https://drive.google.com/file/d/15wpeQlrvOZJzZT7bEMkXu0rhWbpyU4C3/view?usp=sharing>